Lecture 5: Bayesian Inference & Regression

Recall our coin last time

- It was allegedly magic, and we didn't know if it would produce heads at a frequency different from a standard, fair coin

- We used MLE to estimate the true probability, θ^*, of it producing heads on any given flip
  - The chances of it producing tails is $1-\theta^*$
  - The observed frequency/estimated probability of heads is θ
  - The number of heads was represented as k, the number of flips, n
  - Therefore, there are n-k tails

Our “magic” coin – prior probabilities

- A prior probability is a probability assessed *before* you make any relevant observations

- Sounds useless from the definition, but is *fundamentally important* for many, *many* applications
  - Probability of a person at Connecticut College being female?
  - Probability that a coin from my pocket lands on heads when I flip it?
  - Probability that a piece of DNA codes for a gene?
  - Note that you can make observations that may make you change your mind about each of these

An (important) aside: Bayes rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

An (important) aside: Bayes rule

- Example: What is the probability that someone develops lung cancer if they're a smoker?
  - Re-written more statistically: what is the probability that someone develops lung cancer *given* they're a smoker
  - p(lung cancer|smoker) = ?
  - According to Bayes:
    p(lung cancer|smoker) = [p(smoker|lung cancer)*p(lung cancer)] / p(smoker)

  - Note that our prior probability, p(lung cancer) might be what we start with, but we're updating this now that we have additional observations (that this person is a smoker)

An (important) aside: Bayes rule

- Example: What is the probability that someone develops lung cancer given they're a smoker?
  - p(lung cancer|smoker) = [p(smoker|lung cancer)*p(lung cancer)] / p(smoker)
    - p(lung cancer) = 0.06
    - p(smoker) = 0.14
  - Note that it is often fairly easy to get these two probabilities, therefore Bayes rule is useful whenever p(B|A) is easier to figure out than p(A|B)
    - p(smoker|lung cancer) = 0.9

p(lung cancer|smoker) = (0.9 * 0.06) / 0.14

p(lung cancer|smoker) = 0.39 or 39%

Taking the smoking example a bit further

- Example: What is the probability that someone develops lung cancer given they're a smoker?
  - p(lung cancer|**non-**smoker) = [p(**non**-smoker|lung cancer)*p(lung cancer)] / p(**non**-smoker)
    - p(lung cancer) = 0.06
    - p(**non**-smoker) = 0.86
  - Note that it is often fairly easy to get these two probabilities, therefore Bayes rule is useful whenever p(B|A) is easier to figure out than p(A|B)
    - p(**non**-smoker|lung cancer) = 0.1

p(lung cancer|**non-**smoker) = (0.1 * 0.06) / 0.86

p(lung cancer|**non-**smoker) = 0.007 or 0.7%

Our “magic” coin – prior probabilities

- Back to our coin, let’s say a friend insists that the coin isn’t magic and that it’s just a regular coin
  - The *prior probability* will be quite low (say, 10%), as your friend thinks p(heads) is 0.5
  - So, $p(\theta = 0.5) = 0.1$, but $p(\theta = 0.5|D)$ very much *does not*
    - Imagine that your D is HHHHHHHTHHHHHHHHHH, is $p(\theta = 0.5|D)$ 0.5?

$$P(\theta \mid \mathcal{D}) = \frac{P(\mathcal{D} \mid \theta)P(\theta)}{P(\mathcal{D})}$$

Our “magic” coin – p(θ|D)

- What should our hypothesis of p(H) be given our (training) data?
- Well, how about the *Maximum a posteriori* (MAP) probability, which is the mode of the posterior distribution (or the most likely single value)

Our “magic” coin – p(θ|D)

- What should our hypothesis of p(H) be given our (training) data?
- Well, how about the *Maximum a posteriori* (MAP) probability, which is the mode of the posterior distribution (or the most likely single value)
  - That is, select your θ such that p(θ|D) is the highest possible value
  - Similar in concept to MLE, different in that you’re constrained by *prior knowledge* (represented as the prior probability)
  - Note that as your observed knowledge approaches infinity, your confidence in the prior probability goes to zero (the prior is useless if you’re given the answer)
  - This means that Bayes is most useful when you have a good prior and/or a small amount of training data

Our “magic” coin – using numbers

- Say we have the observations HHTHT

$$P(\theta \mid \mathcal{D}) = \frac{P(\mathcal{D} \mid \theta) P(\theta)}{P(\mathcal{D})}$$

- Select θ to be 0.1 (because, why not?)

p(θ|HHTHT) = [p(HHTHT|θ) * p(θ)] / p(HHTHT)

p(θ|HHTHT) α [[0.1*0.1*0.9*0.1*0.9] * ???]

p(θ|HHTHT) α 0.00081

Our “magic” coin – using numbers

- Say we have the observations HHTHT

$$P(\theta \mid \mathcal{D}) = \frac{P(\mathcal{D} \mid \theta) P(\theta)}{P(\mathcal{D})}$$

- Select θ to be 0.6 (because, that’s probably better)

p(θ|HHTHT) α [p(HHTHT|θ) * p(θ)]

p(θ|HHTHT) α [[0.6*0.6*0.4*0.6*0.4] * ???]

p(θ|HHTHT) α 0.03456 * ???

Our “magic” coin – using numbers

- Say we have the observations HHTHT

$$P(\theta \mid \mathcal{D}) = \frac{P(\mathcal{D} \mid \theta) P(\theta)}{P(\mathcal{D})}$$

- Select θ to be 0.5 (to finish off)

p(θ|HHTHT) α [p(HHTHT|θ) * p(θ)]

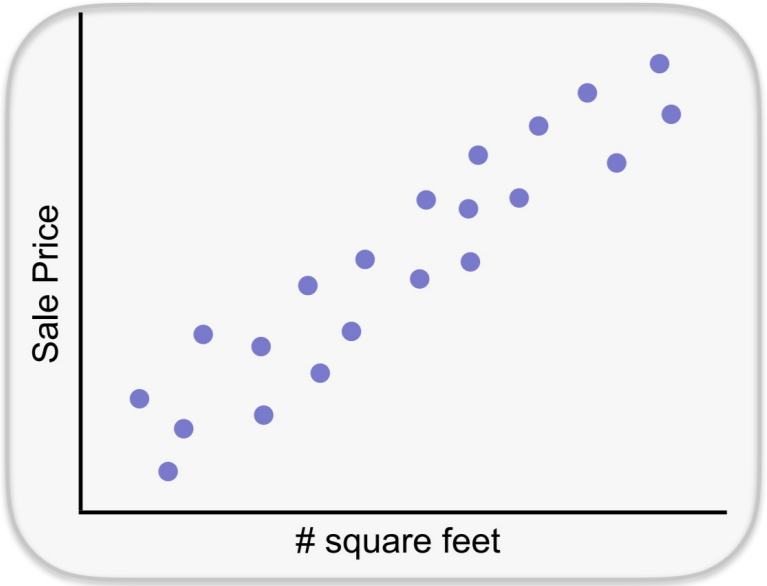
p(θ|HHTHT) α [[0.5*0.5*0.5*0.5*0.5] * ???]

p(θ|HHTHT) α 0.03125 * ???

Lecture 5: Bayesian Inference & Regression

An example regression problem

- Zillow.com – a website that tracks houses for sale & rent
- Given the sales data available, predict what a house will sell for
- y = house sales based on…
- X = square footage, zip code, date of sale, etc., etc.



Training Data:

$$\{(x_i, y_i)\}_{i=1}^{n}$$

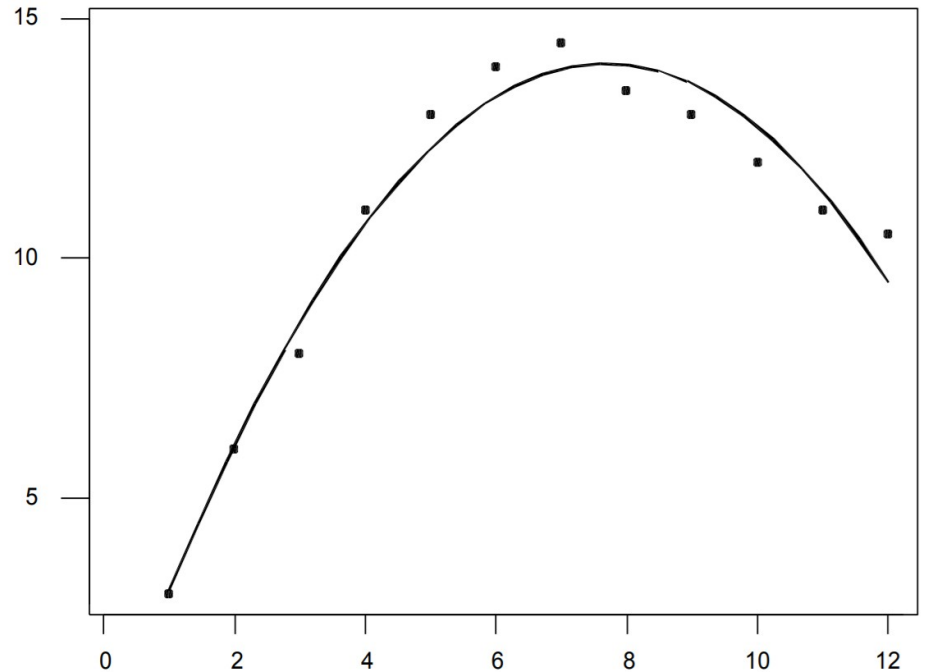
Basis of regression

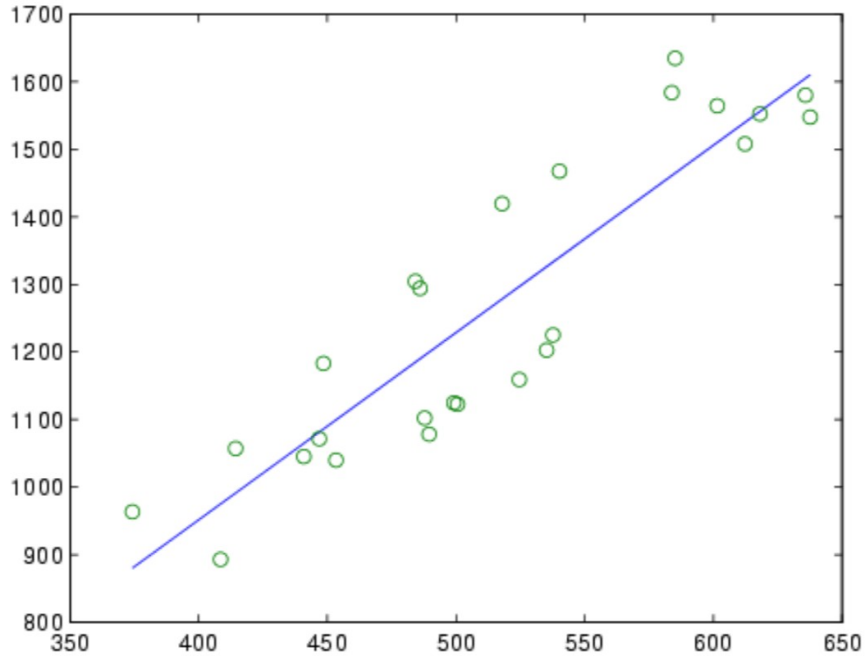
- Useful when you have variables that are related in a ***nondeterministic*** manner (ex: house price and square footage)
- We observe a dependent variable (home price)
- For each dependent variable, we observe predictors (square footage)
- Our goal is to determine the relationship between the dependent variables and predictors

- Note: Dependent variables sometimes called “response(s),” predictors sometimes called “regressor(s)”

Aside: doesn't need to be *linear* regression

- Functions can be (and often are) non-linear
- In this course we will focus on linear regression (other than the next slide) since we're not going to be spending a lot of time on regression in general





Aside: doesn't need to be *linear* regression

- Simple linear regression:

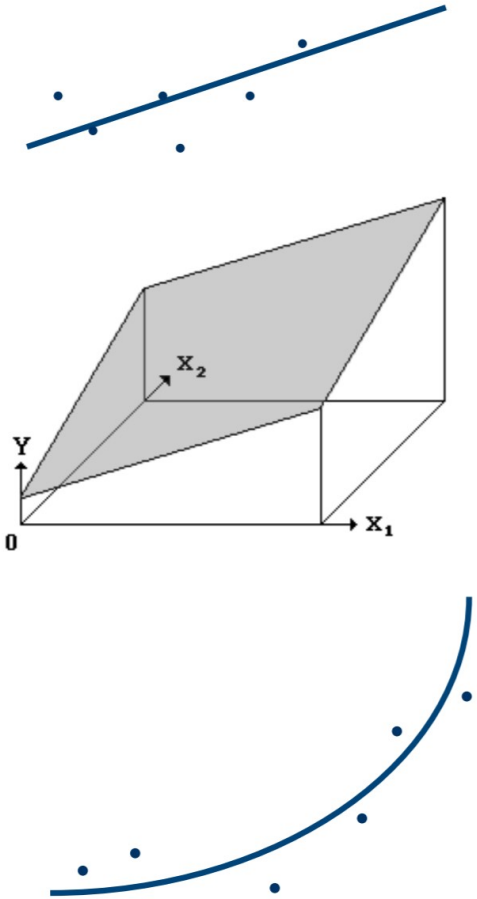
  $Y = B_0 + B_1X + \text{error}$

- Multiple linear regression:

  $Y = B_0 + B_1X_1 + B_2X_2 + \text{error}$

- Polynomial regression:

  $Y = B_0 + B_1X + B_2X^2 + \text{error}$



Linear regression

- Our goal is to find B_0 and B_1 where

  $Y = B_0 + B_1X + \text{error}$

- So we need to find:
  - The intercept, B_0
  - The slope of the line, B_1
- Basic idea (which we'll complicate in a minute):
  - Plot the observations (like a scatter plot)
  - Find the line that best "fits" or follows the plotted points

Estimating regression parameters

- To estimate B_0 and B_1, you want to find the values that minimize the error (specifically, the squared error)
- For observations 1, 2, 3, … i,-1 i, i+1, …, n, minimize:

$$\sum_{i=1}^{n} \left(y_i - (\beta_0 + \beta_1 x_i) \right)^2$$

Least squares (minimizing squared error)

- Minimize the squared error:

$$L = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

The least squares estimators of β_0 and β_1, say, $\hat{\beta}_0$ and $\hat{\beta}_1$, must satisfy

$$\left.\frac{\partial L}{\partial \beta_0}\right|_{\hat{\beta}_0,\hat{\beta}_1} = -2\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\left.\frac{\partial L}{\partial \beta_1}\right|_{\hat{\beta}_0,\hat{\beta}_1} = -2\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0$$

Least squares (minimizing squared error)

The **least squares estimates** of the intercept and slope in the simple linear regression model are

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \tag{11-7}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}} \tag{11-8}$$

where $\bar{y} = (1/n)\sum_{i=1}^{n} y_i$ and $\bar{x} = (1/n)\sum_{i=1}^{n} x_i$.

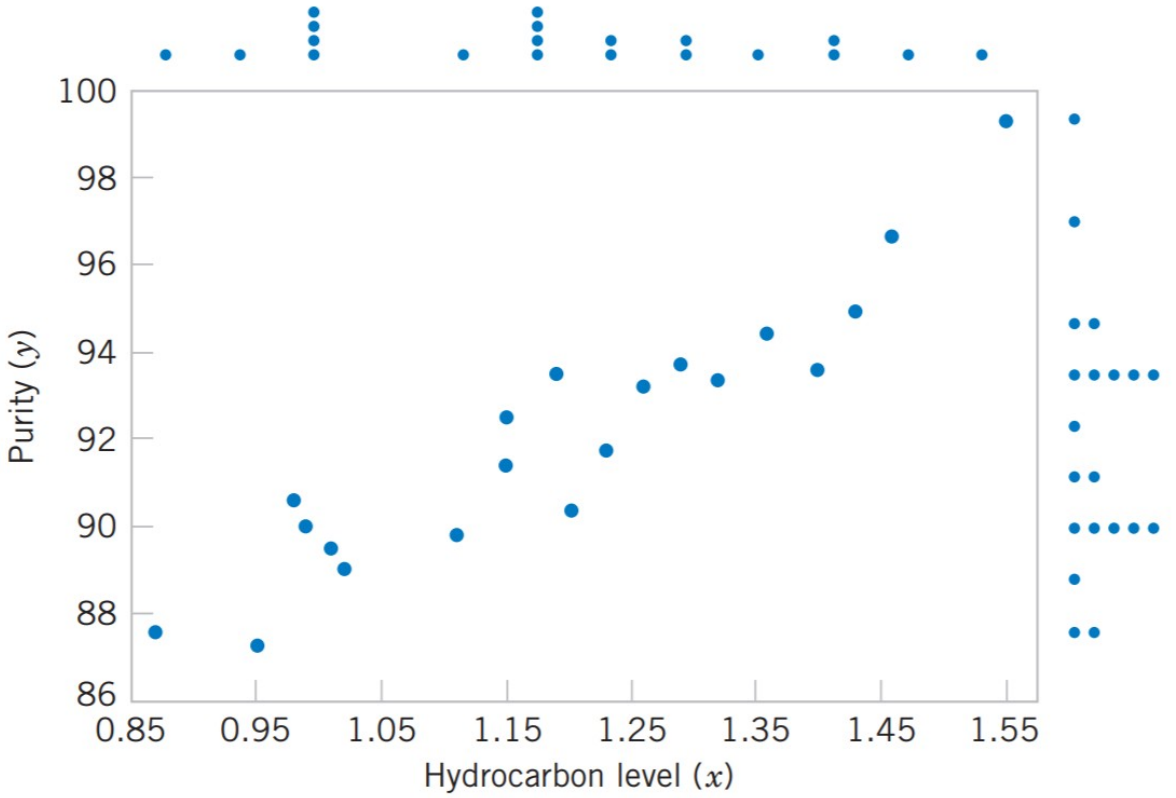
A notation that you might see…

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$$

$$S_{xy} = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}$$

Example: oxygen and hydrocarbon levels

Observation Number	Hydrocarbon Level x (%)	Purity y (%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33



Example: oxygen and hydrocarbon levels

$$n = 20 \quad \sum_{i=1}^{20} x_i = 23.92 \quad \sum_{i=1}^{20} y_i = 1{,}843.21$$

$$\bar{x} = 1.1960 \quad \bar{y} = 92.1605$$

$$\sum_{i=1}^{20} y_i^2 = 170{,}044.5321 \quad \sum_{i=1}^{20} x_i^2 = 29.2892$$

$$\sum_{i=1}^{20} x_i y_i = 2{,}214.6566$$

Example: oxygen and hydrocarbon levels

$$S_{xx} = \sum_{i=1}^{20} x_i^2 - \frac{\left(\sum_{i=1}^{20} x_i\right)^2}{20} = 29.2892 - \frac{(23.92)^2}{20}$$

$$= 0.68088$$

$$S_{xy} = \sum_{i=1}^{20} x_i y_i - \frac{\left(\sum_{i=1}^{20} x_i\right)\left(\sum_{i=1}^{20} y_i\right)}{20}$$

$$= 2{,}214.6566 - \frac{(23.92)(1{,}843.21)}{20} = 10.17744$$

Example: oxygen and hydrocarbon levels

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{10.17744}{0.68088} = 14.94748$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 92.1605 - (14.94748)1.196 = 74.28331$$

Example interpretation

$$\hat{y} = 74.283 + 14.947x$$

$\hat{y} = 89.23\%$ when the hydrocarbon level is $x = 1.00\%$